

Timeformer: A Controlled Diagnostic Study of Temporal Semantic Traceability

Anonymous Author

Institution withheld for review

Abstract. Words change meaning over time, yet most computational models treat token representations as period-independent. We study *temporal traceability*: whether the same surface token, queried at different periods, occupies period-appropriate neighborhoods. A model may improve predictive accuracy with temporal information while still assigning nearly identical neighborhoods across periods; this gap is our focus.

We introduce Timeformer and evaluate it through a controlled design chain. Two baselines bracket the design space: Standard (no temporal information) and Additive (global time vector, no token-time binding). Joint (token and time projected jointly before self-attention) is a direct ablation of Timeformer that removes only the memory component. The full Timeformer extends Joint with causal prototype memory. All models are trained with masked language modeling on a synthetic benchmark with known ground-truth trajectories.

The main finding is that token-time interaction is the effective traceability mechanism. The Joint ablation produces a neighborhood drift of -0.614 for drifting subjects, versus -0.231 for Standard (stable across seeds). Probe-based accuracy gains are modest and seed-sensitive, confirming that geometric diagnostics are more informative than label recoverability. The Memory-Augmented extension shows no robust improvement; oracle analysis identifies mean-prototype aggregation as the failure point.

Keywords: Temporal semantics · Semantic change · Transformer models · Traceability · Diagnostic evaluation

1 Introduction

Words do not preserve fixed meanings across time. In 1900, *broadcast* referred to scattering seeds across a field; today it describes a social media post reaching millions of users. Most language models, however, take no account of when a text was written: a word’s representation depends on its surrounding context, but not on the historical period in which it appeared.

The ability to query a word at a specific period matters beyond technical benchmarks. Whether the shift of *gay* from *cheerful* to its contemporary sense involved movement toward or away from stigmatized terms [3] is an empirical question that requires `gay@1920` and `gay@1980` as directly comparable objects in a shared space.

The dominant computational approach, lexical semantic change detection (LSCD), measures such shifts by training distributional representations on time-sliced corpora and comparing them across periods [6, 7]. One model may associate *broadcast* in 1900 with *sow* and *field*, while another associates it with *signal* and *audience*; change is inferred from the distance between the two spaces. It summarizes the shift as a single similarity score between separately trained models, not as a queryable `token@time` object.

A subtler limitation arises when a single model receives temporal information during training. A model may correctly interpret *virus* as biological infection in 1900 and as malicious software in 2020, improving word-prediction accuracy, while still assigning nearly identical neighborhoods across periods. Prediction accuracy and geometric traceability are separable properties.

We call a representation *temporally traceable* if the same surface token, queried at different periods, occupies neighborhoods consistent with its meaning at that period, making `token@time` a directly inspectable object. This paper introduces Timeformer and evaluates it through a controlled design chain. Standard (no time information) and Additive (global time vector) serve as baselines. Joint (token and time projected jointly before self-attention) is a direct ablation of the full model that removes only the memory component. The full Timeformer extends Joint with causal prototype memory. This chain isolates which form of temporal conditioning produces traceability. We study this in a controlled synthetic benchmark in which the temporal trajectory of each subject is known, giving diagnostic access to target neighborhoods without the confounds of natural historical corpora.

The paper makes four contributions: it formulates temporal traceability as a `token@time` representation problem; introduces the Timeformer design chain; evaluates traceability with diagnostics that inspect subject representations directly (context drift score, contrastive sign-flip rate, and probe-based evaluation); and shows that token-time interaction is the main positive mechanism, while mean-pooled historical memory is better understood as a diagnostic extension.

2 Related Work

Lexical semantic change detection (LSCD) is the dominant computational framing for studying how word meanings shift over time. Diachronic word embeddings [3] made this operational by learning distributional representations for each time slice and measuring movement in embedding space after alignment across independently trained models. The central intuition — that change is visible as a shift in neighborhood structure — is shared by this paper. The gap, however, is architectural: LSCD methods produce a scalar summary of change between two separately trained spaces. They cannot support a `token@time` query, because the two spaces are not jointly parameterized and a token’s representation in one has no direct geometric relationship to its representation in the other. Standardized LSCD benchmarks [6, 7] formalize this scalar-output task but inherit the observational complexity of natural corpora, where frequency shifts

and genre composition confound genuine semantic change. We use a controlled benchmark with planted trajectories to isolate whether traceability emerges from architectural choices rather than corpus properties.

A second line of work incorporates temporal information directly into a single neural model, avoiding separately trained spaces. Dhingra et al. evaluate a time-aware language model on knowledge-intensive QA; Rosin and Radinsky condition a masked language model on document timestamps; Loureiro et al. train period-stratified language models and evaluate on downstream NLP tasks [1, 5, 4]. These models show that temporal conditioning improves predictive accuracy on period-appropriate content. What none of them study is whether temporal conditioning produces *geometrically* traceable representations: whether the same token at different periods occupies genuinely different neighborhoods in a shared space. Accurate period-conditioned predictions are compatible with nearly identical neighborhood structures across times — a dissociation our results confirm and that motivates the diagnostic framework in this paper.

Linear probes are a common tool for testing whether information is encoded in neural representations. A probe trained to predict period labels from a token’s hidden state measures recoverability — whether temporal information is linearly accessible — but does not establish that the same token occupies different neighborhoods at different periods. These are separable properties, and conflating them overstates what probe accuracy can show. We therefore treat probe accuracy as one diagnostic among several and rely on geometric measures of neighborhood movement as the primary evidence for traceability.

Finally, semantic change is not always a clean migration from one meaning to another. Giulianelli et al. show that words can develop two coexisting senses simultaneously, a process better described as bifurcation than drift [2]. Mean-pooled representations collapse both senses into a centroid that faithfully represents neither. The three trajectory classes in our benchmark — stable, drifting, and bifurcating — are designed to test whether temporal conditioning handles this distinction, and the Memory-Augmented extension reveals precisely where mean-pooled prototypes fail.

3 Controlled Temporal Benchmark

The benchmark gives each subject a known temporal trajectory, making it possible to ask directly whether a model’s representation of a subject at t_0 has moved toward the expected neighborhood by t_9 . This distinguishes it from natural-language LSCD benchmarks, where ground truth is often annotated or inferred post-hoc and corpus confounds are hard to separate from genuine semantic change.

The corpus uses a subject–verb–object (SVO) grammar: every sentence consists of one subject, one verb, and one object. Two semantic neighborhoods, N_1 and N_2 , partition the vocabulary of 8 verbs and 8 objects: N_1 uses $\{V1, \dots, V4\}$ and $\{O1, \dots, O4\}$; N_2 uses $\{V5, \dots, V8\}$ and $\{O5, \dots, O8\}$. A sentence generated from N_1 therefore looks like S5 V2 O3, while one from N_2 looks like S5

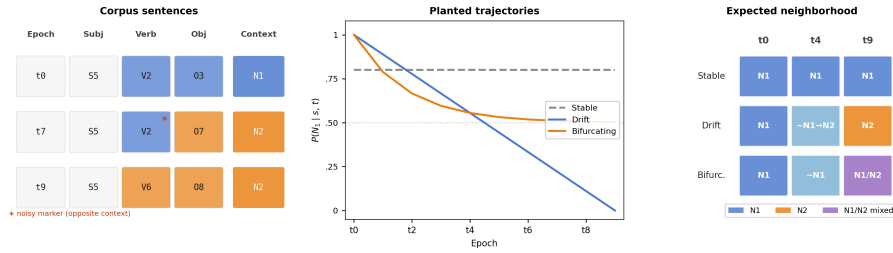


Fig. 1. Benchmark design overview. *Left:* SVO sentences for a drifting subject across three periods; an N_1 verb appearing in an N_2 sentence at $t7$ (marked *) illustrates the deliberate noise in local markers. *Center:* planted $P(N_1 | s, t)$ trajectories for Stable (flat), Drift (monotonic), and Bifurcating (levelling at 0.5) subjects across $t0$ – $t9$. *Right:* expected nearest-neighbor behavior for a traceable representation.

V6 O8; the subject token is the same in both cases, but its co-occurring words differ. Intuitively, the intended meaning of subject s at period t is how often it appears in N_1 sentences; formally, $P(N_1 | s, t)$ is the probability that a sentence is drawn from N_1 . The vocabulary contains 30 subjects; each subject’s trajectory is determined by how $P(N_1 | s, t)$ evolves across the 10 periods $t0$ through $t9$.

Figure 1 (center) shows the three planted trajectory classes. Stable subjects maintain an approximately constant $P(N_1 | s, t)$ throughout. Drift subjects follow a monotonic trajectory from near-exclusive N_1 ($P(N_1 | s, t) \approx 1$ at $t0$) to near-exclusive N_2 ($P(N_1 | s, t) \approx 0$ at $t9$): a drifting subject that appears in sentences like S5 V2 O3 early on will appear in sentences like S5 V6 O8 by the final period. Bifurcating subjects begin in N_1 and move toward a mixed state ($P(N_1 | s, t) \approx 0.5$) where both neighborhoods appear with equal frequency, approximating two coexisting senses. The benchmark contains 10 subjects per class.

Each local marker is drawn from the opposite neighborhood 25% of the time (Figure 1, left), so the model cannot recover the planted neighborhood from co-occurring words alone — it must use subject identity together with the period. The `true_context` label is withheld from training and used only for evaluation.

The benchmark supports several evaluation conditions. The standard split uses the same 75% marker accuracy as training. Two harder splits force the verb, or both verb and object, to come from the opposite context. An ambiguous split sets fidelity to 50%, so local markers carry no reliable signal; only subject identity and period remain. Two additional conditions test temporal behaviour directly and are described in Section 5: the contrastive set presents the same sentence under two different period labels to check whether changing only the period changes the preferred context; the continuation set uses periods held out from training to test generalisation. Results throughout are diagnostic evidence in a controlled setting, not a claim about open-domain historical language.

4 Timeformer Architecture

Timeformer is designed to make a subject representation depend explicitly on the period in which the subject is queried. The full model adds two mechanisms to a standard Transformer encoder. First, it binds token identity and period at the input layer. Second, it lets the current subject consult causal historical prototypes before the encoder. We introduce these mechanisms through a controlled design comparison: two baselines (Standard and Additive), one direct ablation (Token-Time Transformer, labelled Joint in Figure 2), and the full Timeformer.

All models follow the same outer pipeline. A sentence is tokenized as [CLS] S V O [SEP]. Each token w_i is mapped to an input embedding $z_i \in \mathbb{R}^d$. The sequence $(z_1, \dots, z_n)$ is processed by a Transformer encoder, and the final hidden state at the subject position,

$$(h_1, \dots, h_n) = \text{Encoder}(z_1, \dots, z_n), \quad h_s = h_{\text{subj}},$$

is the `token@time` representation inspected by the diagnostics. All models are trained with masked language modeling (MLM): some tokens are masked, and the model must reconstruct them from the remaining tokens and, when available, the period input. What changes across the design comparison is how the subject’s input embedding z_s is built before the encoder sees the sentence.

The temporally informed models use a shared period vector $\tau(t) \in \mathbb{R}^d$. It is a learned projection of sinusoidal features of t/T , where T is the total number of training periods. In plain terms, $\tau(t)$ is the vector representation of the current period. It is distinct from the position encoding p_i , which only tells the Transformer whether a token is in subject, verb, or object position. Additive and Joint differ in how they combine $\tau(t)$ with token identity; the full Timeformer then adds causal memory on top of Joint. Figure 2 summarizes the comparison.

4.1 Baseline: Standard Transformer

The Standard baseline asks whether temporal traceability appears without any period input. Its input embedding is the usual sum of a token embedding and a position encoding:

$$z_i = e(w_i) + p_i.$$

Because no period signal enters the model, the same sentence at t_0 and t_9 produces identical h_s . Standard is the null hypothesis: any temporal behavior must come only from changes in the observed words.

4.2 Baseline: Additive Time-Conditioned Transformer

The Additive baseline asks whether knowing the current period is enough. It adds the same period vector $\tau(t)$ to every token:

$$z_i = e(w_i) + p_i + \tau(t).$$

Figure 2. Controlled design comparison



Fig. 2. Controlled design comparison. **Standard:** Transformer baseline with no temporal signal. **Additive:** temporal baseline that adds the same period vector to every token. **Joint:** direct ablation of Timeformer without memory, using joint token-time projection. **Timeformer:** full model, extending Joint with Prototype Memory and Temporal Attention. New components are highlighted in yellow; dashed grey boxes mark inactive components.

Now S5 V2 O3 at t_0 and t_9 are distinguishable. The subject embedding is $e(\text{S5}) + p_1 + \tau(0)$ at t_0 , and $e(\text{S5}) + p_1 + \tau(9)$ at t_9 . However, the same temporal change is also applied to the verb and object: V2 receives $e(\text{V2}) + p_2 + \tau(t)$, and O3 receives $e(\text{O3}) + p_3 + \tau(t)$. Thus Additive tells the whole sentence which period it belongs to, but it does not explicitly bind period to each token’s identity. If Additive succeeds, global period information is sufficient. If it does not, the next question is whether the period must interact with each token before contextualization.

4.3 Ablation: Token-Time Transformer

The Token-Time Transformer, labelled Joint in Figure 2, is the full Timeformer without the memory component. It asks whether explicit token-time interaction is enough. Instead of adding the same period vector to all tokens, Joint concatenates each token embedding with $\tau(t)$ and projects the pair:

$$z_i = W[e(w_i); \tau(t)] + p_i,$$

where $e(w_i), \tau(t) \in \mathbb{R}^d$ and $W \in \mathbb{R}^{d \times 2d}$. The difference from Additive is visible in the example. At t_9 , the period vector is combined separately with S5, V2, and O3. The model computes $W[e(\text{S5}); \tau(9)]$, $W[e(\text{V2}); \tau(9)]$, and $W[e(\text{O3}); \tau(9)]$ as separate inputs. The same period can therefore affect the subject differently

from the verb or object. For traceability, the key case is the subject: S5 at t_0 receives $W[e(\text{S5}); \tau(0)] + p_1$, while S5 at t_9 receives $W[e(\text{S5}); \tau(9)] + p_1$. The subject starts from an epoch-specific input before self-attention, making $h_s(\text{S5}, t_0)$ and $h_s(\text{S5}, t_9)$ free to move in different semantic directions.

4.4 Full Model: Timeformer

The full Timeformer asks whether causal history helps beyond token-time interaction. It starts exactly like Joint: for a current sentence containing S5 at t_7 , the model first builds token-time input embeddings for [CLS] S5 V O [SEP]. It then modifies only the subject embedding before the Transformer encoder sees the sentence.

The additional information comes from Prototype Memory. Timeformer stores one prototype vector per subject and past period. A prototype $m(s, t)$ is the average representation of subject s across training sentences from period t . The memory is causal: at t_7 , the model may consult $\{m(\text{S5}, 0), \dots, m(\text{S5}, 6)\}$, but not prototypes from t_7 or later.

Temporal attention turns this list of past prototypes into a history summary for the current subject. Let z_s be the current Joint subject embedding and let M_s be the matrix of available causal prototypes, one row per past period. The attention operation uses z_s as the query and M_s as keys and values:

$$r_s = \text{Attn}(z_s, M_s, M_s).$$

The vector r_s summarizes the parts of the subject’s history that are most relevant to the current period. A learned scalar gate g then decides how much this history summary should change the current subject embedding:

$$\tilde{z}_s = z_s + g \cdot (\text{LN}(z_s + r_s) - z_s).$$

Here, LN denotes layer normalization, and g is a learned scalar gate initialized at zero so that Timeformer reduces to Joint at initialization. Finally, $\tilde{z}_s$ replaces z_s in the input sequence; verb and object embeddings are unchanged, so the subject is the only token that enters the encoder informed by causal prototype history.

5 Evaluation Protocol and Diagnostics

All models in the comparison chain are trained with masked language modeling over the synthetic SVO corpus. The planted `true_context` label is withheld from training and used exclusively for post-hoc evaluation: it tests whether the learned subject representation reflects the planted temporal trajectory. The primary evaluation target is h_s , the final hidden state at the subject position, because subjects are the lexical items whose meanings are designed to drift, remain stable, or bifurcate over time.

We use four diagnostics, each targeting a distinct aspect of temporal representation.

D1 – Linear probe. Asks whether h_s encodes the planted context label at all. We train a linear classifier on h_s from the standard test split to predict `true_context` and evaluate it on each temporal split. High accuracy indicates that period-appropriate context is linearly separable in the representation, but does not reveal whether that separation shifts systematically over time.

D2 – Context drift score. Asks whether the neighborhood of a subject shifts in the expected direction across periods. For each period and subject class, we retrieve the $k = 10$ nearest neighbors of h_s by cosine similarity (excluding the query itself) and record the proportion assigned to N_1 . For drifting subjects, this proportion should decrease monotonically from t_0 to t_7 ; for stable subjects it should remain near its initial value; for bifurcating subjects it should decrease less steeply, as the representation splits between neighborhoods rather than migrating from one to the other.

D3 – Contrastive diagnostic. Asks whether the period label alone drives representational change. We present the same sentence with two different period labels and compute the sign-flip rate: the proportion of pairs for which swapping the period reverses the nearest-neighbor assignment, from N_1 -dominant to N_2 -dominant or vice versa. A model that ignores the period input produces a sign-flip rate of zero; a model that responds to it produces a rate above zero.

D4 – Continuation diagnostic. Asks whether the learned temporal mapping generalizes beyond training. We evaluate on held-out periods t_8 – t_9 , excluded from MLM training. A model that has internalized the direction of change, rather than memorizing period-specific patterns, should maintain coherent neighborhood assignments in these unseen periods.

The four diagnostics are interpreted through a fixed ablation chain. Additive vs. Standard isolates the effect of adding any temporal signal. Joint vs. Additive tests whether that signal must interact with token identity (per-token gating) rather than being applied globally as a uniform shift. Timeformer vs. Joint tests the contribution of causal prototype memory, the mechanism most expected to affect D3 and D4.

For the Memory-Augmented extension, we include three controls. In the *shuffled-memory* control, each subject receives prototypes drawn from a different subject, disrupting identity-specific history while preserving the memory pathway. In the *no-history* control, the memory path is present but empty. In the *oracle-memory* diagnostic, learned prototypes are replaced by fixed vectors encoding the true historical $P(N_1 \mid s, t)$ trajectory. The extension passes the evaluation criterion if it outperforms both the Token-Time Transformer and the two controls; the oracle diagnostic then indicates how much headroom remains relative to a perfect memory signal.

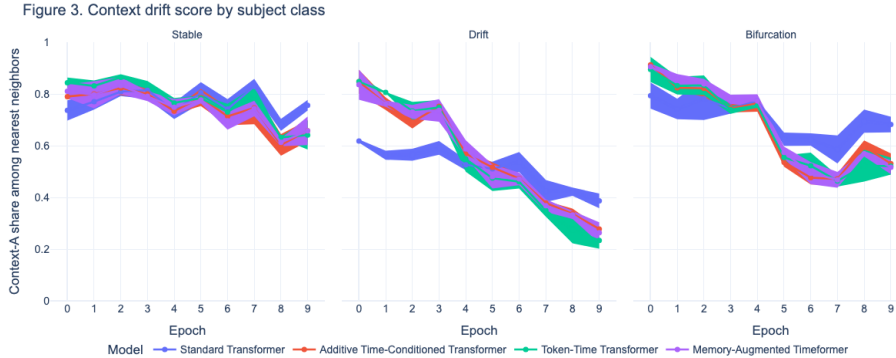


Fig. 3. Context drift score across periods. Temporal baselines move drifting subjects from N_1 toward N_2 more strongly than the Standard Transformer, with the Token-Time Transformer showing the largest transition. Stable and bifurcating subjects provide class-specific controls.

6 Results

We report results over three training runs. Unless otherwise noted, values are means over seeds, with the standard deviation shown in parentheses when space allows. The main finding is that temporal conditioning produces a much clearer temporal movement in representation space than the Standard Transformer, with token-time interaction giving the strongest drifting-subject transition. The Memory-Augmented extension is informative as a diagnostic, but it does not provide a robust improvement over the Token-Time Transformer.

6.1 Temporal Neighborhood Movement

Figure 3 and Table 1 show the context drift score across periods. For drifting subjects, the Standard Transformer drops from 0.618 to 0.387 ($\Delta = -0.231$). The Additive baseline shows a stronger transition ($\Delta = -0.570$), and the Token-Time Transformer strengthens it further, from 0.847 to 0.233 ($\Delta = -0.614$). The gap between the Token-Time Transformer and Standard is the primary geometric evidence for temporal traceability: token-time interaction causes the same subject to occupy substantially different neighborhoods at different periods.

The class-specific curves serve as controls (Table 1). Stable subjects show much smaller movement and no coherent N_1 -to- N_2 trend. Bifurcating subjects show intermediate drift consistent with a trajectory toward a mixed state, not a complete transition. The geometric diagnostic distinguishes all three trajectory classes without relying on probe accuracy alone.

Table 1. Context drift score at the first and last periods. Values are N_1 proportions among nearest neighbors; $\Delta = t9 - t0$.

Class	Model	t0	t9	Δ
Stable	Standard	0.736	0.756	+0.020
	Additive	0.790	0.658	-0.132
	Joint	0.843	0.641	-0.203
	Timeformer	0.810	0.658	-0.153
Drift	Standard	0.618	0.387	-0.231
	Additive	0.847	0.278	-0.570
	Joint	0.847	0.233	-0.614
	Timeformer	0.836	0.263	-0.573
Bifurcation	Standard	0.794	0.682	-0.112
	Additive	0.913	0.531	-0.382
	Joint	0.895	0.521	-0.374
	Timeformer	0.905	0.517	-0.388

6.2 Probe and Contrastive Diagnostics

The linear probe results show modest accuracy gains that are consistent with, but weaker evidence for, temporal traceability than the neighborhood analysis. On the ambiguous split, where local verb and object markers are uninformative, the Standard Transformer reaches 0.577 accuracy, the Additive baseline reaches 0.573, and the Token-Time Transformer reaches 0.599. The corresponding mean delta for token-time interaction is +0.026, but it is seed-sensitive: the three runs give +0.053, +0.005, and +0.021. Probe accuracy alone is insufficient to establish temporal traceability: it measures recoverability of the planted label, not the geometry of temporal movement.

The contrastive sign-flip diagnostic gives a different view. The Standard Transformer has a sign-flip rate of 0.000, as expected. The Additive baseline averages 0.690 while the Token-Time Transformer averages 0.517, a counterintuitive ordering. The decoupling arises because sign-flip counts threshold crossings in individual contrastive pairs, whereas drift score measures the full trajectory magnitude. A global additive offset can move representations substantially and generate many threshold crossings, but it applies the same period signal to every token. Token-time interaction produces a slightly stronger drifting-subject trajectory while making the period signal token-dependent; it does not necessarily cross the threshold in more individual contrastive pairs. The sign-flip rate confirms that both time-conditioned models respond to the period input; the neighborhood trajectory (D2) remains the more informative measure of how the geometry changes across periods.

6.3 Memory Diagnostics

The Memory-Augmented Timeformer does not produce a robust gain over the Token-Time Transformer. On continuation, both models average 0.766, and the

Table 2. Main traceability diagnostics. Accuracy columns report linear-probe accuracy on h_s ; flip is contrastive sign-flip rate.

Model	Drift	Flip	Ambig.	Cont.
Standard	-0.231	0.000	0.577	0.719
Additive	-0.570	0.690	0.573	0.754
Joint	-0.614	0.517	0.599	0.766
Timeformer	-0.573	0.506	0.599	0.766

Table 3. Memory diagnostics. Continuation (D4) is linear-probe accuracy on held-out periods $t8-t9$; flip is the contrastive sign-flip rate (D3).

Model	Cont.	Flip	Δ vs. Joint
Joint (reference)	0.762	0.448	–
Timeformer learned	0.775	0.448	+0.013 cont.; +0.000 flip
Timeformer oracle	0.754	0.586	–0.008 cont.; +0.138 flip

per-seed memory deltas are +0.013, +0.005, and -0.018 . The oracle diagnostic clarifies the failure mode: oracle memory improves contrastive sign flip (0.586 vs. 0.448 for the Token-Time Transformer) but not continuation (0.754 vs. 0.762). Perfect historical prototypes can affect contrastive temporal behavior, but do not solve the continuation objective under the current MLM setup.

7 Discussion and Conclusion

This paper asked whether a Transformer-style encoder can expose a temporally traceable representation — one in which the same lexical item, queried at different periods, occupies neighborhoods that reflect its intended temporal context. The answer, in the controlled benchmark used here, is conditional: explicit token-time interaction is necessary. A Standard encoder and a globally time-conditioned encoder both fail to produce coherent subject-level trajectories. An encoder that conditions each token embedding on time before self-attention does produce such trajectories, with a neighborhood drift for drifting subjects more than twice that of the Standard baseline (-0.614 vs. -0.231).

The central methodological lesson is that probe accuracy and neighborhood geometry measure different things. A model can use period information to improve predictions without producing a representation in which the same lexical item occupies meaningfully different neighborhoods at different periods. The evaluation protocol contributes a method for separating these two properties: geometric diagnostics are necessary alongside label recoverability, as each captures a distinct aspect of temporal traceability.

This work explores a family of temporal conditioning designs of increasing complexity. In this benchmark, the Token-Time Transformer is the most effective

instantiation: per-token conditioning allows the same period to affect different tokens differently, producing coherent subject-level trajectories where a global offset does not. The negative result for the Memory-Augmented extension is specific to mean-pooled prototypes in this controlled setting and should not be read as a general verdict on causal memory. The oracle diagnostic confirms this: perfect historical prototypes improve the contrastive sign-flip rate by +0.138 over the Token-Time Transformer, showing the memory pathway is useful in principle. The failure of learned prototypes is a failure of aggregation: for bifurcating subjects, which develop two coexisting senses in later periods, the mean prototype is a centroid between two clusters, representing neither well.

Several limitations bound these conclusions. The benchmark is synthetic: subjects follow planted trajectories in a small controlled vocabulary, and results may not transfer directly to natural corpora with noisier or more gradual semantic change. The evaluation covers 30 subjects over three seeds, which is sufficient to establish direction but not to quantify effect sizes precisely. The MLM training objective optimizes token prediction rather than trajectory geometry, so traceability emerges as a side effect rather than a training target. Finally, the diagnostic suite was co-designed with the architecture family, which limits its role as an independent evaluation for models outside this design space.

Designing memory mechanisms that preserve the internal structure of historical representations — rather than collapsing them to a single prototype — is the most direct next step. Evaluating token-time interaction on natural corpora with attested semantic change is the necessary path toward claims beyond this controlled setting.

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Dhingra, B., Cole, J.R., Eisenschlos, J.M., Gillick, D., Eisenstein, J., Cohen, W.W.: Time-aware language models as temporal knowledge bases. *Transactions of the Association for Computational Linguistics (TACL)* **10**, 257–273 (2022)
2. Giulianelli, M., Del Tredici, M., Fernández, R.: Analysing lexical semantic change with contextualized word representations. In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*. pp. 3960–3973 (2020)
3. Hamilton, W.L., Leskovec, J., Jurafsky, D.: Diachronic word embeddings reveal statistical laws of semantic change. In: *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (ACL)*. pp. 1489–1501 (2016)
4. Loureiro, D., Barbieri, F., Neves, L., Anke, L.E., Camacho-Collados, J.: TimeLMs: Diachronic language models from Twitter. In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL): System Demonstrations*. pp. 251–260 (2022)
5. Rosin, G., Radinsky, K.: Time masking for temporal language models. In: *Proceedings of the 15th ACM International Conference on Web Search and Data Mining (WSDM)*. pp. 833–841 (2022)

6. Schlechtweg, D., McGillivray, B., Hengchen, S., Dubossarsky, H., Tahmasebi, N.: SemEval-2020 task 1: Unsupervised lexical semantic change detection. In: Proceedings of the 14th International Workshop on Semantic Evaluation (SemEval). pp. 1–23 (2020)
7. Tahmasebi, N., Borin, L., Jatowt, A.: Survey of computational approaches to lexical semantic change. Computational Approaches to Semantic Change pp. 1–91 (2021)